

Height of Random Binary trees

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1 Introduction

Our aim is to investigate the height of uniformly random rooted trees with vertex set $[n]$, where the number of children of each node is specified in advance. We look at the work of Addario-Berry and Donderwinkel [1], summarizing and reproving aspects of it in the specific case of full binary trees, we will justify our decision shortly, by describing how this case can be used to stochastically upper bound the height of more general trees. We describe a bijection, first introduced by Foata and Fuchs [2], between trees and sequences, on which our calculations are based. Then, we use this bijection to perform a warm-up calculation, looking at the height of the so-called ‘least labeled leaf’ and use this result to bound the height of an arbitrary leaf. Next, we give a proof of the base case of Lemma 13 from [1] in the binary case, and explain how this key result fits into the larger picture of tail bounds of heights of random trees. We conclude with a result showing that the height of the leaves of a full binary tree stochastically dominate the height of the internal vertices.

To begin, let’s specify what we mean by a uniformly random rooted tree. Since we are looking at uniformly random *rooted* trees, our element t is really a tuple (t, r) where t is a tree structure, and r the vertex marked as the root. For the rest of this report, however, we will simply refer to this tuple as t .

In particular, we will look at trees with fixed degree sequences. A degree sequence $d = (d_1, d_2, \dots, d_n)$ is a sequence of non-negative integers, where for all $i \in [n]$, d_i specifies the number of children of the vertex i . Thus, it holds that $\sum_{i \in [n]} d_i = n - 1$.

For a degree sequence $d = (d_1, \dots, d_n)$, $\mathcal{T}_d$ is the set of all trees t on $[n]$ where vertex i has degree d_i , $\forall i \in [n]$. When we say ‘ T is a uniformly random tree with degree sequence d ’ we mean that T is sampled uniformly at random from $\mathcal{T}_d$, which we denote by $T \in_u \mathcal{T}_d$.

1.1 It suffices to consider binary trees

Before we begin the main part of the report, we give a brief justification for considering only binary trees. In particular, we will only consider *full* binary trees, that is to say, binary trees where all internal vertices have degree two. We claim that bounding the height of such trees allows us to provide an upper bound on the height of trees with general degree sequences. There are two logical steps to this argument.

The first step in our justification is that among all random trees for the same given n and with the same number of degree-one vertices, sub-binary trees (trees with maximum degree 2) have the stochastically largest heights—up to a technical detail that is required to address potential parity issues. This is proven as Corollary 12 by Addario-Berry and Donderwinkel in [1]. Thus, we can derive bounds for all trees by simply considering sub-binary trees. It remains for us to argue that looking at *full* binary trees is sufficient.

Let d be a degree sequence with $n_1 > 0$ entries equal to one, and let $n' = n - n_1$. Then, to obtain a uniformly random element of $\mathcal{T}_d$, we can do the following: Let d' be d with all the degree one vertices removed, so d' has length n' . Then, sample $T' \in_u \mathcal{T}_{d'}$. Subdivide the edges of T' according to a uniformly random $(n' - 1)$ -composition of n_1 , and label the new vertices according to an independent uniformly random permutation of the set $\{i \in [n] : d_i = 1\}$. Letting T be the resulting tree, it turns out that $T \in_u \mathcal{T}_d$. Fairly standard computations about uniformly random compositions show that the number of subdivisions on a fixed edge of T' is typically only $O(n_1/n')$ and moreover the number of subdivisions on a long path in T' is concentrated enough that typically all root to leaf paths P in T' are only subdivided $O(|P|n_1/n')$ times. This is made precise and quantitative in [1, Lemma 15].

The main question, which we focus on for the remainder of the report, then becomes bounding the height of full binary trees. From this point forward, we will simply refer to full binary degree sequences as binary degree sequences, since it is all we will be discussing.

2 A bijection between trees and sequences

In this section, we give a bijection, first introduced by Foata and Fuchs [2], between rooted trees on $[n]$ and certain sequences in $[n]^{n-1}$. The version we give is described by Addario-Berry and Donderwinkel [1], and is obtained from the Foata-Fuchs bijection by specializing to trees with a given degree sequence. The bijection is presented for general degree sequences, however, in certain places we note how it looks specifically for binary degree sequences.

First, to make our degree sequences more convenient to deal with, we focus on compressed degree sequences—this just means a degree sequence where all non-zero entries appear before all of its zero entries. We can turn any degree sequence into a compressed degree sequence by simply moving the zero entries to the end. Formally, let $d = (d_1, \dots, d_n)$ be a degree sequence. We define another degree sequence $d' = (d'_1, \dots, d'_n)$ as follows: let m be the number of non-zero degree entries in d . For a binary degree sequence, $m = \frac{n-1}{2}$. We define $d'_1, \dots, d'_m$ as the first m non-zero entries of d , in the order they appear in d . We then set $d'_{m+1} = \dots = d'_{n-1} = 0$.

This presents a natural bijection between $\mathcal{T}_d$ and $\mathcal{T}_{d'}$ (note that this is a bijection between sets of trees, not between sets of degree sequences). Given $t \in_u \mathcal{T}_d$ construct a tree $t' \in \mathcal{T}_{d'}$ as follows: relabel the non-leaf vertices of t as $1, \dots, m$ and the leaves of t as $m + 1, \dots, n - 1$. In both cases, this is done in increasing order of their original labels. Then, since the bijection consists simply of relabeling, it provides a coupling (T, T') of uniformly random elements of $\mathcal{T}_d$ and $\mathcal{T}_{d'}$ such that T and T' have the same height. Thus, any tail bound we prove on the height of a uniformly random tree in $\mathcal{T}_{d'}$ applies identically to the height of a uniformly random tree in $\mathcal{T}_d$. From this point on, it suffices to consider only compressed degree sequences.

Let $d = (d_1, \dots, d_n)$ be a compressed degree sequence. As before, m is the number of non-

zero entries. Denote the number of leaves in d as $n_0 = n_0(d) = n - m$ (in the binary case we can simply write $n_0 = \frac{n+1}{2}$). We define a set of sequences, contained in $[n]^{n-1}$:

$$\mathcal{S}_d := \{(v_1, \dots, v_{n-1}) : |\{k : v_k = i\}| = d_i, \forall i \in [n]\}$$

In words, $\mathcal{S}_d$ is the set of all tuples of length $n - 1$, which contains d_i copies of i for each $i \in [n]$. For a sequence $v = (v_1, \dots, v_{n-1}) \in \mathcal{S}_d$, we define the location of a repeat of v to be any $j \in \{2, \dots, n - 1\}$ such that $v_j = v_i$ for some $i < j$. We now state a bijection between $\mathcal{S}_d$ and $\mathcal{T}_d$, as it appears in [1].

Bijection between $\mathcal{S}_d$ and $\mathcal{T}_d$

- Let $j(0) = 1$, and let $j(1) < j(2), \dots < j(n_0 - 1)$ be the locations of the repeats in the sequence v . Let $j(n_0) = m + n_0 = n$
- For $i = 1, \dots, n_0$ let P_i be the path $(v_{j(i-1)}, \dots, v_{j(i)-1}, m + i)$
- Let $t(v) \in \mathcal{T}_d$ have root v_1 and edge set the union of the edge sets of the paths $P_1, P_2, \dots, P_{n_0}$.

This bijection can be viewed as constructing the tree t from the sequence $v(t)$ in an iterative manner. As bookkeeping, we must keep a list of the leaves, in increasing order of label. Then, to construct the tree, we begin by placing the first entry in the sequence as the root of the tree. Next, we place each successive node in the sequence in a path from the root, until the first time we come across an entry in the sequence that we have already seen. At this point, we cap off our path with the first (i.e. smallest labeled) leaf. Then, we begin a new path, starting from the repeated vertex. Once again, we build this path out until we see a repeat in our sequence, at which point we cap the path with the next leaf, and begin building again from the repeated vertex. For each occurrence of i in v , a path is constructed from i to a leaf, so vertex i has degree $|\{k : v_k = i\}| = d_i$ in $t(v)$. Therefore $t(v) \in \mathcal{T}_d$ and the map $t : \mathcal{S}_d \rightarrow \mathcal{T}_d$ is well-defined.

We now state the inverse of the bijection. Fix a tree $t \in \mathcal{T}_d$. Let $S_0 = \{r(t)\}$ be the set consisting of the root of t . For $1 \leq i \leq n_0$ we define P_i to be the path from S_{i-1} to the node $m + i$ in t , and set $S_i = S_{i-1} \cup P_i$. Let P_i^* be P_i excluding its final node $m + i$. Then, let $v(t) = P_1^*, \dots, P_{n_0}^*$. One can check that $v(t) \in \mathcal{S}_d$, and moreover that $v : \mathcal{T}_d \rightarrow \mathcal{S}_d$ is the inverse of $t : \mathcal{S}_d \rightarrow \mathcal{T}_d$. Hence, both maps are well-defined bijections. It follows that if $T \in_u \mathcal{T}_d$, then $v(T) \in_u \mathcal{S}_d$.

3 Bounding the height of a leaf

3.1 Height of the least labeled leaf

As a warm-up, we calculate the height of the leaf whose label is smallest among all leaves in a random binary tree. We call this the ‘least labeled leaf’ and will denote it ℓ . That is, we fix a binary degree sequence $b = (b_1, \dots, b_n)$ and set $\ell = \min\{i \in [n] : b_i = 0\}$. For this calculation, we will make use of the bijection we described in the last section and think of a tree $t \in \mathcal{T}_b$ as being constructed from a sequence $v = v(t) \in \mathcal{S}_b$.

What is the least labeled leaf in this context? It is the first leaf we add as we go through the iterative process to construct the tree from v . Specifically, this means the least labeled leaf occurs the first time we encounter a repetition in the entries of v . So, we can think of bounding the height of ℓ as bounding the probability of having a string of non-repeating entries at the beginning of v .

For a tree t and a vertex i , we write $\text{ht}_t(i)$ for the *height* of i in t , which is the number of edges on the path in t from i to the root. For $T \in_u \mathcal{T}_b$ and $k \geq 0$, we’ll bound $P(\text{ht}_T(\ell) > k)$.

Letting $V = (V_1, \dots, V_n) = v(T) \in_u \mathcal{S}_d$, this is the same things as bounding the probability that each of the first $k + 1$ entries of V are all different. We can write out the expression for this probability as:

$$P(V_2 \neq V_1) \cdot P(V_3 \neq V_1, V_3 \neq V_2 | V_1 \neq V_2) \cdots P(V_{k+1} \neq V_i, \forall i < k+1 | V_i \neq V_j, \forall i \neq j \leq k+1).$$

Think of our process as drawing from a bag which contains two of each internal vertex; since there are $(n-1)/2$ internal vertices, when you make two copies of each, the sequence ends up with length $n-1$. Then, the number of items we are drawing from decreases by one each time. Also, since we are conditioning on having no prior repetitions, the number of items we can't draw increases by one each time. Thus, we can write our expression as:

$$\left(1 - \frac{1}{n-2}\right) \cdot \left(1 - \frac{2}{n-3}\right) \cdots \left(1 - \frac{k}{n-(k+1)}\right).$$

This is the main idea behind our proof. Now let's state the theorem and use this intuition to formally prove it.

Theorem 1. *Let $b = (b_1, \dots, b_n)$ be a binary degree sequence, and ℓ the least labeled leaf in b . Let $T \in_u \mathcal{T}_b$. Then for all $x \geq 0$,*

$$P(\text{ht}_T(\ell) > \lceil x\sqrt{n} \rceil) \leq e^{-x^2/2}.$$

Proof. As we stated above, for all $k \geq 0$,

$$P(\text{ht}_T(\ell) > k) = \left(1 - \frac{1}{n-2}\right) \cdot \left(1 - \frac{2}{n-3}\right) \cdots \left(1 - \frac{k}{n-(k+1)}\right).$$

Now we will use some standard tricks to clean this up into a nicer bound. □

$$\begin{aligned} \prod_{i=1}^k \left(1 - \frac{i}{n-(i+1)}\right) &\leq \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k}{n}\right) \\ &\leq e^{-1/n} \cdots e^{-k/n} \\ &= \exp\left(-\sum_{i=1}^k \frac{i}{n}\right) \\ &= \exp\left(-\frac{1}{n} \cdot \frac{k(k+1)}{2}\right) \\ &\leq \exp\left(\frac{-k^2}{2n}\right). \end{aligned}$$

So we get that

$$P(\text{ht}_T(\ell) > k) \leq \exp\left(\frac{-k^2}{2n}\right).$$

Now, let $k = \lceil x\sqrt{n} \rceil$, where we are required to take the ceiling since our argument to derive our bound only works for positive integers k . The above expression becomes

$$P(\text{ht}_T(\ell) > \lceil x\sqrt{n} \rceil) \leq \exp\left(\frac{-\lceil x\sqrt{n} \rceil^2}{2n}\right) \leq e^{-x^2/2}.$$

This is exactly what we wanted. We see that $\text{ht}_T(\ell)/\sqrt{n}$ has subgaussian tails. □

The distribution of the height of the least leaf turns out to converge to a known random variable, so let's look at that quickly as well. First, rescale our result from the theorem to get

$$P\left(\frac{\text{ht}_T(\ell)}{\sqrt{n}} > \frac{\lceil x\sqrt{n} \rceil}{\sqrt{n}}\right) \leq e^{-x^2/2}$$

But in fact, the inequalities in the preceding calculation are asymptotically tight when k is of order $\sqrt{n}$. Furthermore, $\lceil x\sqrt{n} \rceil/\sqrt{n} \rightarrow x$ as $n \rightarrow \infty$. Hence, for all $x \geq 0$, as $n \rightarrow \infty$,

$$P\left(\frac{\text{ht}_T(\ell)}{\sqrt{n}} > x\right) \rightarrow e^{-x^2/2}.$$

That is, $\text{ht}_T(\ell)/\sqrt{n}$ converges in distribution to a standard Rayleigh random variable as $n \rightarrow \infty$.

3.2 Height of an arbitrary leaf

We can translate our bound on the height of the least labeled leaf to a bound on the height of an arbitrary leaf. In fact, we will prove something slightly stronger: the distribution of the heights of any two leaves are the same. For this result, we forget about the bijection with sequences, and simply think in terms of random binary trees.

Theorem 2. *Let b be a binary degree sequence and fix any two leaves ℓ_1 and ℓ_2 . Let $T \in_u \mathcal{T}_b$. Then $\text{ht}_T(\ell_1) \stackrel{d}{=} \text{ht}_T(\ell_2)$; in other words, for all $x \geq 0$, $\mathbf{P}(\text{ht}_T(\ell_1) = x) = \mathbf{P}(\text{ht}_T(\ell_2) = x)$.*

Proof. Consider the transformation $\pi: [n] \rightarrow [n]$ which switches the labels of ℓ_1 and ℓ_2 , and leaves the labels of all other nodes fixed. Formally,

$$\pi(v) = \begin{cases} \ell_1 & \text{if } v = \ell_2 \\ \ell_2 & \text{if } v = \ell_1 \\ v & \text{else} \end{cases}$$

By extending this definition such that $\pi(t)$ simply applies π to all vertices in $t \in \mathcal{T}_b$, π is a permutation on $\mathcal{T}_b$. Thus, for all $t \in \mathcal{T}_b$, we have $\mathbf{P}(T = t) = \mathbf{P}(\pi(T) = t)$. That is, $T \stackrel{d}{=} \pi(T)$. At the same time, by definition of π , we have $\text{ht}_t(\ell_1) = \text{ht}_{\pi(t)}(\ell_2)$ for all $t \in \mathcal{T}_b$. Therefore, for all $x \geq 0$,

$$\mathbf{P}(\text{ht}_T(\ell_1) = x) = \mathbf{P}(\text{ht}_{\pi(T)}(\ell_2) = x) = \mathbf{P}(\text{ht}_T(\ell_2) = x).$$

And we conclude that $\text{ht}_T(\ell_1) \stackrel{d}{=} \text{ht}_T(\ell_2)$. □

In the next section, we go over the steps necessary to bound the height of a random binary tree. A first attempt to do so might be to use the union bound to translate our bound on the height of a leaf to a bound on the height of the whole tree. Unfortunately, this gives us a weaker bound than what we derive in the next section. Indeed, the union bound is tight when events are independent, but the height of leaves is not—and this fact will help us.

To provide some brief intuition before we dive into the next section, recall our proof of the height of the least labeled leaf. It relied on showing the sequence $v \in \mathcal{S}_b$ is not likely to have a long string at the beginning without any repetitions. Now consider the second leaf in the sequence. Right away, it starts off with more potential nodes who, upon being drawn, would result in a repetition. This makes it less likely to have another long sequence without repetitions. The idea that as the tree gets ‘bushier’ as we continue building it, it gets harder to have a long string of entries without repetitions, is the intuition behind our proof in the next section.

4 Bounding the height of a tree

In this section, we look at bounding the height of random binary trees. We adapt part of the proof of a key result [1, Lemma 13] to the binary case, then sketch how it fits into the big picture.

To start, we will need a bit more notation. We list the non-leaf vertices $\{1, \dots, m = \frac{n-1}{2}\}$ in the order they appear when constructing a tree t from $v(t)$ as $(i(1), \dots, i(m))$. Note that, since we only consider binary degree sequences (with no degree one entries), $i(x)$ is the first node in the vector $v(t)$ such that the subtree of t consisting of all nodes of $t_{i(x)}$ and all their children have strictly more than $\lceil x \rceil$ leaves.

4.1 Lemma 13

Lemma 13 from [1] is stated below in the binary case.

Lemma 3. *Fix any binary degree sequence $b = (b_1, \dots, b_n)$. Then, for all $0 \leq x \leq y$ and for any integer $c \geq 0$, if $T \in \mathcal{T}_b$ then*

$$P(\text{dist}(v_{i(x)}, T_{i(x)}) > c) \leq \left(1 - \frac{x}{n-1}\right)^c$$

Proof. We assume b is compressed (and in particular $b_n = 0$). Throughout the proof, we take $T \in_u \mathcal{T}_b$ and let $V = v(T)$, so $V \in_u \mathcal{S}_b$.

Fix any vector $(i(1), \dots, i(j))$ of distinct elements of $[n]$ such that $j \geq y$, and let $\mathcal{S}_b(i(1), \dots, i(j))$ be the set of vectors $v \in \mathcal{S}_b$ such that, writing $t = t(v)$,

$$(i_t(1), \dots, i_t(j)) = (i(1), \dots, i(m))$$

In words, $\mathcal{S}_b$ is the set of vectors with the same first j non-leaf nodes (added in the same order) as t .

For $V \in_u \mathcal{S}_b$, the ordering of the integers $\{1, \dots, n\}$ by their first appearance depends on degree. Since all the elements of $(i(1), \dots, i(m))$ are non-leaves, they all have degree two, so

$$\begin{aligned} P((i_T(1), \dots, i_T(j)) = (i(1), \dots, i(j))) &= \prod_{k=1}^j \frac{d_i(k)}{n-1-d_{i(1)}-\dots-d_{i(k-1)}} \\ &= \prod_{k=1}^j \frac{2}{n-1-2(k-1)} \end{aligned}$$

To derive this expression, think of constructing $(i_T(1), \dots, i_T(j))$ by drawing from a pool containing two of each non-leaf vertex (where two is the degree of each non-leaf vertex), and upon drawing a vertex, removing all copies of it from the pool.

Using the above equality, we can write,

$$|\mathcal{S}_b(i(1), \dots, i(j))| = \binom{n-1}{2, \dots, 2} \cdot \prod_{k=1}^j \frac{2}{n-1-2(k-1)}$$

Where $2, \dots, 2$ represents a list of $\frac{n-1}{2}$ twos. The binomial expression represents the total number of elements in $\mathcal{S}_b$, and the product is the fraction of them that fulfill the requirements to be in $\mathcal{S}_b(i(1), \dots, i(j))$.

Fix $\ell < y$. Let $b^\ell = (b_1^\ell, \dots, b_{n-1}^\ell)$ where $b_{i(\ell)}^\ell = b_{i(\ell)} - 1$ and $b_i^\ell = b_i \forall i \in [n-1] \setminus \{i(\ell)\}$. Observe that b^ℓ is a legal degree sequence, since $d_n = 0$ so $\sum_{i \in [n-1]} b_i^\ell = n-2$. We now wish to consider the subset of $\mathcal{S}_b(i(1), \dots, i(j))$ where $i(y)$ is a (direct) child of $i(\ell)$. We can describe this formally by taking the subset of $\mathcal{S}_b^\ell(i(1), \dots, i(j))$ of $\mathcal{S}_b(i(1), \dots, i(j))$ consisting of vectors v where the first instance of $i(y)$ in v is an immediate successor of the second instance of $i(\ell)$. We claim

that $\mathcal{S}_b^\ell(i(1), \dots, i(j))$ is in bijection with the set of vector $\mathcal{S}_{b^\ell}(i(1), \dots, i(j))$. $\mathcal{S}_{b^\ell}(i(1), \dots, i(j))$ is defined in the same manner as before, namely as the set of vectors $v' \in \mathcal{S}_{b^\ell}$ such that

$$(i(1, v'), \dots, i(j, v')) = (i(1), \dots, i(j))$$

It is reasonable to expect a bijection between these two sets, since in terms of counting the number of distinct trees, $i(\ell)$ having one fewer child is equivalent to fixing a specific vertex as the child of $i(\ell)$. We formalize this bijection as follows. Fix $v \in \mathcal{S}_b^\ell(i(1), \dots, i(j))$ and let v' be the vector obtained by deleting the immediate predecessor of $i(y)$, which we recall is some instance of $i(\ell)$. However, since the immediate predecessor of $i(y)$ is the second instance of $i(\ell)$, we have that $(i(1, v), \dots, i(j, v)) = (i(j, v'), \dots, i(j, v'))$. To see the inverse of the bijection, take v' and insert an entry with value $i(\ell)$ directly preceding the first instance of $i(y)$ in v' . Thus, the sizes of $\mathcal{S}_b^\ell(i(1), \dots, i(j))$ and $\mathcal{S}_{b^\ell}(i(1), \dots, i(j))$ are the same. Then, by the same computation we used for the size of $\mathcal{S}_b(i(1), \dots, i(j))$ we get

$$\begin{aligned} |\mathcal{S}_b^\ell(i(1), \dots, i(j))| &= |\mathcal{S}_{b^\ell}(i(1), \dots, i(j))| \\ &= \binom{n-2}{2, \dots, 2} \left(\prod_{k \in [j] \setminus \{\ell\}} \frac{2}{n-2-2(k-1)} \right) \left(\frac{1}{n-2-2(k-1)} \right) \end{aligned}$$

Where the $2, \dots, 2$ represents a list of $\frac{n-1}{2} - 1$ twos this time, as $d_{i(\ell)}^\ell = d_{i(\ell)} - 1 = 2 - 1 = 1$. Now, we let $E(y, \ell)$ be the event that the first instance of $i(y, V)$ in V is an immediate successor of the second instance of $i(\ell, V)$. It follows that

$$\begin{aligned} \mathbf{P}(E(y, \ell) \mid (i_T(1), \dots, i_T(j)) = (i(1), \dots, i(j))) \\ &= \frac{|\mathcal{S}_b^\ell(i(1), \dots, i(j))|}{|\mathcal{S}_b(i(1), \dots, i(j))|} \\ &= \frac{1}{n-1} \cdot \prod_{k=1}^j \frac{n-1-2(k-1)}{n-1-2(k-1)} \\ &> \frac{1}{n-1} \end{aligned}$$

We note that if $E(y, \ell)$ occurs then $i_T(\ell)$ is the parent of $i_T(y)$ in $T = t(V)$. Then, on the event $\{(i_T(1), \dots, i_T(j)) = (i(1), \dots, i(j))\}$ the tree $T_{i(x)}$ has vertices $i(1), \dots, i(x)$. Thus,

$$\begin{aligned} \mathbf{P}(\text{par}_T(i_T(y)) \in t_{i(x)}(V) \mid (i_T(1), \dots, i_T(j)) = (i(1), \dots, i(j))) \\ &\geq \sum_{\ell=1}^x \mathbf{P}(E(y, \ell) \mid (i_T(1), \dots, i_T(j)) = (i(1), \dots, i(j))) \\ &\geq \frac{x}{n-1} \end{aligned}$$

Finally, by summing over $(i_T(1), \dots, i_T(j))$ this proves the case $c = 1$ of the lemma. Indeed, we actually proved something slightly stronger for the case $c = 1$: that the necessary bound holds even conditionally given $(i_T(1), \dots, i_T(j))$.

The full lemma is proven by arguing inductively. Unfortunately, to prove the induction step as [1] does, an induction hypothesis applying solely to binary degree sequences is not strong enough. Thus, to see the remainder of the proof, we refer the reader to the original version found in [1]. $\square$

4.2 The big picture

After proving Lemma 13, [1] extend the result to obtain Corollary 14. Corollary 14 bounds the probability that the difference in heights between the trees $T_{i(y)}$ and $T_{i(x)}$ is large, that is to say, it bounds the probability of the distance from $T_{i(x)}$ to any vertex added between $i(x)$ and $i(y)$ being large, whereas Lemma 13 just looks at the probability of the distance between $T_{i(x)}$ and $i(y)$ being large. To do so, they use the fact that if some vertex $i(z)$ is distance c from $T_{i(x)}$, then there must be at least $(1-p)c$ vertices v along the path from $T_{i(x)}$ to $i(z)$ for which $\text{dist}(T_{i(x)}, v)$ is greater than $\lceil pc \rceil$.

By combining Corollary 14 with a telescoping sum bound, [1] are able to bound the height of random trees $T \in_u \mathcal{T}_d$ for a degree sequence d , where $n_1(d) = 0$. A bound for general degree sequences (no restriction on $n_1(d)$) is then derived in [1], Lemma 15]. This is done using the argument of subdividing edges according to a uniformly random composition, which we describe in subsection 1.1. Finally, Lemma 15 is a key input to proving [1], Theorem 6], which bounds the height of random trees. The proof has many technical details, but the broad idea is to incrementally construct T in layers, so as to control how much is added to the height at a time. [1], Lemma 15] is used to bound how much the height increases by during the steps of this incremental construction.

5 Height of an Arbitrary Internal Vertex

In our last sections, we show that in a full binary tree, the height of the leaves stochastically dominates the height of the internal vertices. By combining this with our results from Section 3.1 and 3.2, we can achieve an upper bound on the height of an arbitrary internal vertex.

To begin, we present a pairing between internal vertices and leaves, where each internal vertex is paired to a unique leaf of greater height. This will allow us to prove our desired result.

Theorem 4. *Let t be a binary tree. View t as a subdivision of a full binary tree, and denote that full binary tree by t' . Then, there exists an injection ϕ from the internal vertices of t' to its leaves, such that*

1. $\text{ht}_{t'}(v) \leq \text{ht}_{t'}(\phi(v))$ for all internal vertices v of t'
2. $\text{Im}(\phi) = L \setminus \{\ell\}$ where L is the set of leaves of t' (which is the same as the set of leaves of t) and ℓ is a leaf of maximum height in t .

Proof. We induct on n , the number of internal vertices of t' .

Base case ($n = 1$) : Our tree t' is a root r with two children that are leaves in t' . By assigning $\phi(r)$ to be its child with minimum height in t , our conditions are satisfied.

Induction step ($n - 1 \implies n$) : Let u be an internal vertex of maximum height in t' . Then, u has two children which are leaves in t' . Let $\phi_1(u)$ map to u 's child w with minimum height in t .

Let P be the set of vertices that subdivide the edge uw . Now, let s denote $t \setminus (\{w\} \cup P)$. Then, s can be viewed as the subdivision of some full binary tree s' with $n - 1$ internal vertices. We may apply our induction hypothesis, to obtain a injection ϕ_2 .

We now define

$$\phi(x) = \begin{cases} \phi_1(x) & \text{if } x = u \\ \phi_2(x) & \text{if } x \text{ is an internal vertex of } s' \end{cases}$$

And we claim that ϕ is our desired injection. Let V denote the internal vertices of s' . Note that $u \notin V$ and $V \cup \{u\}$ is equal to the internal vertices of t' , so ϕ is well defined on the proper

domain. Further, it is an injection, since w is not in s' . Next, note that property (1) is clearly inherited from ϕ_1 and ϕ_2 . Finally, note that for any leaf ℓ in s' , its height in s is the same as the height that the corresponding vertex of s' had in s . Then, since the other child of u in s' (the child that is not w) has height in t at least as great as w , a leaf ℓ of maximum height in s' is also a leaf of maximum height in t' . Thus, property (2) must be satisfied by ϕ . $\square$

$\square$

We use this to prove that the height of leaves stochastically dominates the height of internal vertices in each fixed $t \in \mathcal{T}_b$. After, we will extend this to apply for $T \in_u \mathcal{T}_b$.

Corollary 1. *Let t be a full binary tree. Then the height of the leaves of t stochastically dominates the height of its internal vertices. Formally, for V a random internal vertex and L an independent random leaf,*

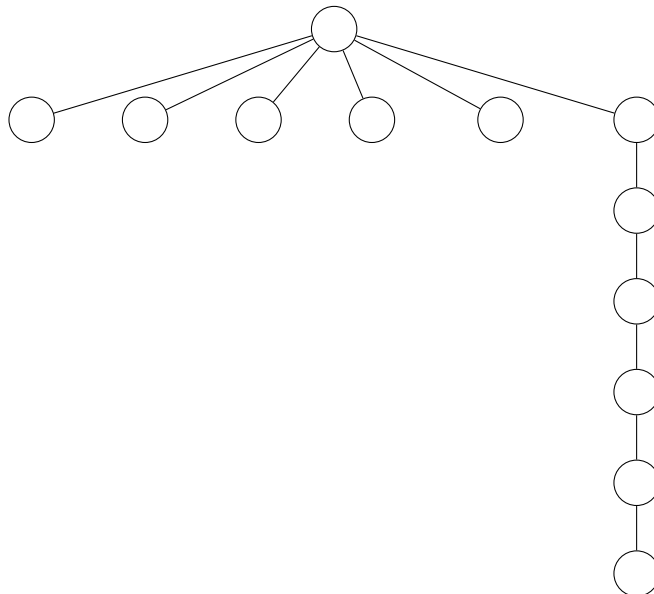
$$P(\text{ht}_t(V) > x) \leq P(\text{ht}_t(L) > x)$$

Proof. Let V be sampled randomly from the internal vertices of t , let L be sampled randomly from the leaves of t , and let L' be sampled randomly from $\text{Im}(\phi)$.

We will prove our theorem by providing two couplings: one between V and L' such that $\text{ht}_t(V) \leq \text{ht}_t(L')$, and the second coupling between L' and L such that $\text{ht}_t(L') \leq \text{ht}_t(L)$. It will then follow that $\text{ht}_t(V) \preceq_{\text{st}} \text{ht}_t(L') \preceq_{\text{st}} \text{ht}_t(L)$, which is our desired result.

For the first coupling, let V be sampled randomly from the internal vertices and let $L' = \phi(V)$. Since ϕ is an injection from the internal vertices of t to the leaves of t , it is a bijection between the internal vertices of t and $\text{Im}(\phi)$. Thus $L' \in_u \text{Im}(\phi)$, so we have a valid coupling. By property 1 of ϕ , we have $\text{ht}_t(V) \leq \text{ht}_t(\phi(V)) = \text{ht}_t(L')$ as required. $\square$

The result proven in the corollary above may seem rather intuitive. It may even be tempting to think the same result can be proven for arbitrary trees, not just binary trees; however, this is not the case. To see this, consider a tree which is rooted on a vertex of high degree. The root has many children which are leaves, and one child off of which is a long path. Then, all of the leaves except one have height 1, whereas all of the internal vertices save two have height greater than one. Clearly, the height of the leaves will not dominate that of the internal vertices in this tree, since almost all of the leaves are at distance 1 from the root, and almost all of the internal vertices have distance at least 2 from the root. Below, we give an example of one such tree.



This shows that we cannot hope to generalize our corollary to apply to general trees. With this observation out of the way, we now extend our corollary to hold for $T \in_u \mathcal{T}_b$, thereby proving our main result.

Corollary 2. *Let b be a binary degree sequence. Let $T \in_u \mathcal{T}_b$; Let V a random internal vertex; and let L a random leaf, all drawn independently. Then,*

$$P(ht_t(V) > x) \leq P(ht_t(L) > x)$$

In words we have that the height of the leaves of T stochastically dominates the height of the internal vertices of T .

Proof.

$$\begin{aligned} P(ht_t(V) > x) &= \sum_{T \in \mathcal{T}_b} P(ht_t(V) > x) \cdot P(T = t) \\ &\leq P(ht_t(L) > x) \cdot P(T = t) \\ &= P(ht_t(L) > x) \end{aligned}$$

Where the inequality on the second line follows from the previous corollary □

References

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